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Alzheimer Disease Classification based on brain volumes trajectories

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Abstract

This report presents a nonparametric analysis for a project in *Nonparametric Statistics*. We investigate the classification of subjects into cognitively normal (CN), mild cognitive impairment (MCI), and Alzheimer's disease (AD) using MRI-derived brain volumes in a nonparametric framework. In a first pipeline, longitudinal trajectories are summarized into regional atrophy rates via quantile regression and mapped into a low-dimensional projection-depth space, where multiple classifiers (DD- α , conformal prediction sets, k NN, and RBF SVM) achieve moderate performance, suggesting substantial overlap between groups. Motivated by this limitation, we then adopt a functional-data perspective and represent each subject by multivariate volume curves over age, applying functional depth (modified band depth) classifiers and depth-based logistic models. Overall, results indicate that the main discriminative signal is not a sharp location shift but a gradual and heterogeneous departure from a normative CN pattern, with MCI displaying the strongest overlap with both CN and AD.

The source code of the entire project is available at

- https://github.com/MatteoZanetti775/NPS_project_alzheimer

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1 Problem statement

1.1 Dataset presentation

The dataset is provided by the **Alzheimer’s Disease Neuroimaging Initiative (ADNI)**, a longitudinal multisite study designed to develop clinical, imaging, genetic, and biochemical biomarkers for the early detection and tracking of Alzheimer’s Disease (AD).

The study involves 1176 subjects who initially joined the project. These individuals present heterogeneous clinical trajectories: some remain stable throughout the follow-up, while others experience a diagnostic conversion.

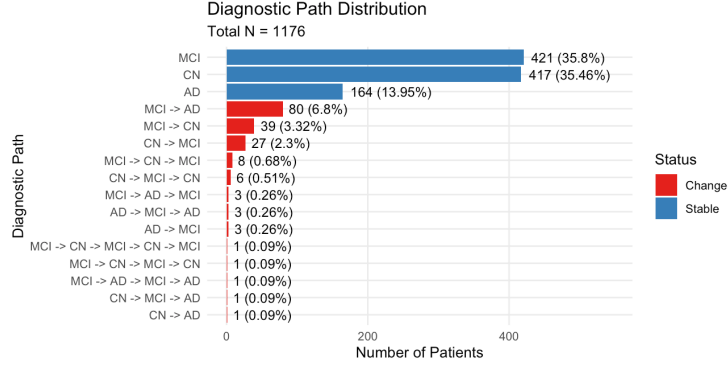


Figure 1: Data representation and longitudinal structure across different clinical statuses: **CN** (Cognitively Normal), **MCI** (Mild Cognitive Impairment), and **AD** (Alzheimer’s Disease).

A key statistical challenge of this dataset is its unbalanced nature. As summarized in Table 1, follow-up paths vary significantly in both duration and frequency, with observations irregularly spaced in time.

Table 1: Summary of the ADNI longitudinal dataset.

Quantity	Value
Number of subjects	1176
Number of visits	3080
Visits per subject (range)	1 – 9
Follow-up time (months)	0 – 132
Age range (years)	55.0 – 91.5
Number of brain regions (volumes)	117

Variables: The dataset includes clinical metadata and high-dimensional neuroimaging features. The volumetric variables represent the volume of specific anatomical structures extracted from magnetic resonance images (MRI).

Table 2: Main variables used in the analysis.

Variable	Description
RID	Unique subject identifier.
month	Time in months from the baseline visit.
age	Age of the subject at the specific visit.
DIAGNOSIS	Clinical diagnosis at the current visit: CN , MCI and AD .
GENOTYPE	Genetic information representing the APOE genotype ($\epsilon 2/\epsilon 3/\epsilon 4$). The presence of the $\epsilon 4$ allele is the primary genetic risk factor for AD.
DXDEP	Binary indicator (0/1) for the presence of depressive symptoms.
Brain Volumes	117 regional volumes (e.g. Right Hippocampus , Left Amygdala) measured in mm^3 .

1.2 Stakeholder Analysis and Clinical Impact

Targeted at clinicians and researchers, this framework provides an objective tool to monitor disease progression beyond traditional cognitive tests. Its main benefits include:

- **Objective Diagnosis:** Acts as a data-driven second opinion when cognitive results like the MMSE are unclear.
- **Robust Monitoring:** Captures global brain changes over time, reducing the measurement errors found in single MRI scans.
- **Better Care Planning:** Helps hospitals prioritize resources for patients whose brain trajectories show the fastest decline.

1.3 Research Question

The core objective of this project is to address the following research question:

Is it possible to develop a robust, non-parametric framework based on longitudinal brain volume trajectories to improve the classification of patients into the three diagnostic stages: CN, MCI, and AD?

To answer this, the analysis is built upon two distinct methodological strands that summarize the longitudinal structure of the dataset into patient-specific features:

1. **Atrophy Rates Estimation:** Instead of treating each visit as an independent observation, we leverage the temporal progression. For each subject and brain region, we estimate an *atrophy rate* representing the volume trend over time. This reduces the multi-visit history into a p -dimensional vector of regional slopes, allowing us to classify subjects based on their speed of neurodegeneration.
2. **Functional Depth Mapping (MBD):** We treat the longitudinal visits as continuous trajectories (Functional Data). Using the Modified Band Depth (MBD), we summarize the entire sequence of visits into one depth value. This index measures how "central" a patient's brain aging profile is compared to a reference group (e.g., healthy controls).

Based on these two representations, we develop and compare different classification strategies, including Support Vector Machine, k-NN, Conformal classification, Max-Depth Classifiers and Logistic Regressions with genetic covariates.

2 Classification based on Atrophy Rates

2.1 Data Preprocessing and Cleaning

The initial dataset consisted of $N = 1176$ subjects. To ensure the reliability of the longitudinal analysis, a rigorous cleaning pipeline was applied:

- **Missing Values:** Visits with missing values in the primary diagnostic labels or brain volumes were removed.
- **Longitudinal patients only:** Since both the estimation of *atrophy rates* and *functional depths* (MBD) require a temporal trend, subjects with only a single visit were excluded. This led to the removal of 299 patients.
- **Final Cohort:** The final analysis was conducted on 877 subjects.
- **Train/Test Split Strategy:** To evaluate the predictive power on disease progression, we perform 5-fold CV on the cohort made by the 703 stable patients.

2.2 Estimation of Atrophy Rates

We explored different statistical frameworks to summarize the longitudinal history of 117 brain regions into a single p -dimensional vector of *atrophy rates*.

Models investigated

- **Quantile Regression (QR - M0):** A local, non-parametric approach to estimate the median trend ($\tau = 0.5$). Specifically, the quantile regression model, robust to noise in MRI segmentation, is fitted independently for each subject, using years from baseline as the sole covariate.
- **Linear Mixed-Effects (LME - M1, M2, M3):** Modeling both population-level fixed effects and subject-specific random intercepts and slopes. Specifications include covariates such as *APOE* genotype, age, head size, and depression (DXDEP).
- **Robust Bayesian LME:** A hierarchical model implemented via `brms` using a Student-t likelihood and normal priors. While theoretically superior for subjects with very few observations, this approach was tested only on a representative subset of 5 brain regions due to its extreme computational cost. The time-intensive nature of MCMC sampling for all 117 regions made it unfeasible for the full high-dimensional feature extraction.

Model Selection and Performance Comparison. The estimation of atrophy rates was validated through a 10-fold cross-validation on the Mean Absolute Error (MAE). We compared the individual Quantile Regression (M0) against three specifications of Linear Mixed-Effects (LME) models.

As shown in the analysis, the individual non-parametric approach (M0) outperformed all hierarchical models by a significant margin, achieving an average MAE of 24.50 compared to the 409.09 of the best-performing LME (M3). The paired Wilcoxon signed-rank test yielded $V = 0$ and $p < 0.005$ for all comparisons, providing strong evidence that local median-based estimation is more robust to the inherent noise of longitudinal MRI data.

While the Bayesian LME offered a robust probabilistic framework, it was ultimately excluded from the final pipeline. The decision was driven by the trade-off between its prohibitive computational demand for the full 117-region dataset and the fact that the simpler QR (M0) already provided a sufficiently high level of precision and robustness. Consequently, Model 0

was selected as the optimal framework, where for each subject i and region j , the volume y_{ij} is modeled as:

$$y_{ij} = \beta_{0,i} + \beta_{1,i}t_{ij} + \epsilon_{ij}, \quad i = 1, \dots, I; \quad j = 1, \dots, n_i \quad (1)$$

with t_{ij} representing the years from baseline. The final features used for classification are the Annual Percentage Changes (APC), computed as:

$$\text{Atrophy Rate}_{\%}^{(i)} = \frac{\hat{\beta}_{1,i}}{\hat{\beta}_{0,i}} \times 100 \quad (2)$$

This normalization effectively ensures that the neurodegeneration signal is comparable across the cohort regardless of the subjects' initial brain size.

2.3 Assessing Regional Sensitivity to Neurodegeneration/Regional Validation: Identifying Key Atrophy Markers

To validate the biological signal, we performed **Univariate Permutational Independence Tests** with Benjamini-Hochberg correction. The identification of the Inferior Lateral Ventricle, Thalamus and Entorhinal Cortex as top biomarkers confirms the anatomical validity of our features. Large mean differences between groups and bootstrap 95% confidence intervals ($R = 1000$) demonstrate the magnitude and stability of these differences. These robust trends provide a reliable basis for the subsequent multivariate classification.

2.4 Dimensionality Reduction through Depth

After estimating subject-specific atrophy rates via quantile regression, we obtain for each subject a p -dimensional feature vector $x_i \in \mathbb{R}^p$, where each component corresponds to the atrophy rate of a brain region.

Working directly in this high-dimensional space can be unstable and hard to interpret, especially in the presence of:

- potential outliers and heterogeneous patterns,
- weak or unknown parametric assumptions on class-conditional distributions,
- dimensionality-related issues (noise accumulation, poor geometry in high- p settings).

For this reason, we adopt a *depth-based* approach: instead of classifying subjects by raw atrophy magnitudes, we classify them by *how typical they are* with respect to each diagnostic group. The entire analysis is organized as a two-stage pipeline:

1. **Common representation step:** map each subject to a low-dimensional *depth space*;
2. **Classification step:** apply one (or more) classifiers directly in depth space.

2.5 Depth-space representation

Let $\mathcal{G} = \{\text{CN}, \text{MCI}, \text{AD}\}$ be the set of diagnostic groups. For each group $g \in \mathcal{G}$ we compute a statistical depth function $D_g(\cdot)$, which measures the *centrality* of a subject relative to the distribution of group g :

- **high depth** $\Rightarrow$ subject is typical/central for that group,
- **low depth** $\Rightarrow$ subject is atypical/peripheral for that group.

In this work, depth is computed via *projection depth*. Since projection depth relies on robust univariate summaries along many directions, we first standardize the features with robust statistics and then compute class-wise depth functions. Each subject is finally represented by three depth coordinates:

$$z_i = (D_{\text{CN}}(\tilde{x}_i), D_{\text{MCI}}(\tilde{x}_i), D_{\text{AD}}(\tilde{x}_i)) \in \mathbb{R}^3, \quad (3)$$

transforming the original p -dimensional space into a common, geometrically interpretable 3D space. This representation is used as the foundation for all the classification methods explored in this project.

2.5.1 Robust standardization

Atrophy-rate features may contain extreme values. Therefore, each feature is standardized using the sample median (location) and the median absolute deviation (MAD) (scale). For feature $j = 1, \dots, p$:

$$m_j = \text{median}\{x_{1j}, \dots, x_{nj}\}, \quad \text{MAD}_j = \text{median}\{|x_{1j} - m_j|, \dots, |x_{nj} - m_j|\},$$

and we define the robustly standardized data:

$$\tilde{x}_{ij} = \frac{x_{ij} - m_j}{\text{MAD}_j}. \quad (4)$$

This step reduces sensitivity to outliers and ensures features are comparable on a common scale.

2.5.2 Projection depth and DD-plot mapping

For each group $g \in \mathcal{G}$, projection depth is computed using only subjects belonging to group g as reference. Intuitively, projection depth quantifies how far a point lies from the bulk of a distribution when examined across multiple one-dimensional projections.

A standard definition is:

$$D_g(x) = \frac{1}{1 + O_g(x)}, \quad O_g(x) = \sup_{\|u\|=1} \frac{|u^\top x - \text{median}(u^\top X_g)|}{\text{MAD}(u^\top X_g)}, \quad (5)$$

where X_g denotes the empirical sample from class g and the supremum is taken over all unit directions u . In practice, the supremum is approximated numerically by evaluating a large set of directions.

After computing $D_{\text{CN}}(\cdot)$, $D_{\text{MCI}}(\cdot)$ and $D_{\text{AD}}(\cdot)$, each subject is mapped to:

$$z_i = (D_{\text{CN}}(\tilde{x}_i), D_{\text{MCI}}(\tilde{x}_i), D_{\text{AD}}(\tilde{x}_i)),$$

yielding a 3D DD-plot representation: subjects typical of a group tend to have higher depth relative to that group. Figure 2 shows the resulting 3D DD-plot in depth space, where each point corresponds to one subject and colors indicate the true diagnostic label.

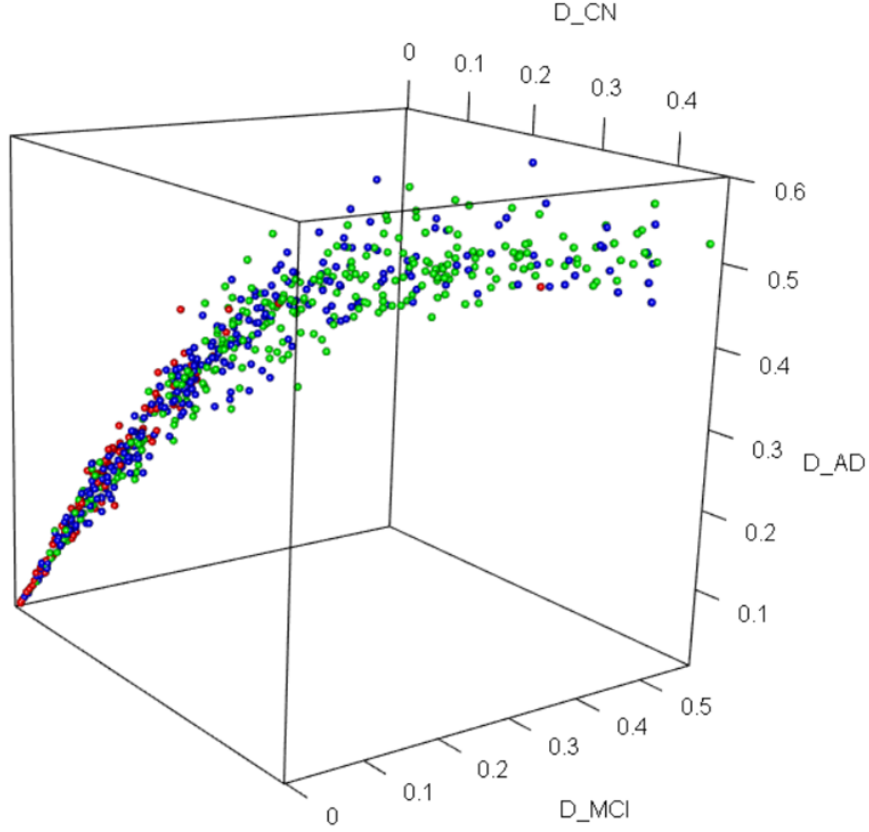


Figure 2: 3D DD-plot (depth-space representation) based on projection depth. Each point represents a subject with coordinates $(D_{CN}, D_{MCI}, D_{AD})$; colors denote the true diagnostic group (CN/MCI/AD).

2.6 Method 1: DD- α classifier in projection depth space

Overview. Method 1 applies the DD- α classification rule directly to the depth vectors $\{z_i\}$. Since preprocessing and depth computation are common to all methods, the key component here is the classifier operating in the 3D depth space.

DD- α classification. The DD- α procedure constructs decision boundaries in depth space by exploiting the geometry of the mapped points. It does not require linear separability and does not impose parametric assumptions on the class distributions, as it relies on relative centrality rather than raw feature magnitudes.

Performance evaluation. Performance is assessed using stratified 5-fold cross-validation: class proportions are preserved in each fold, the classifier is trained on 4 folds and tested on the remaining fold. As a global summary, we report the cross-validated mean accuracy:

$$\text{Mean accuracy (5-fold CV)} = 0.5401. \quad (6)$$

Class-wise performance is summarized by the aggregated cross-validated confusion matrix reported in Table 3.

Table 3: Confusion matrix for DD- α (5-fold cross-validation). Rows indicate predicted labels and columns indicate true labels.

Predicted \ True	AD	CN	MCI
AD	28	4	13
CN	5	168	113
MCI	71	103	167

Interpretation. Overall accuracy is moderate ($\approx 54\%$), reflecting the intrinsic difficulty of separating the three diagnostic groups using atrophy-rate information alone. Table 3 shows that most errors involve confusion with the MCI group, which is expected given its intermediate clinical profile. In particular, a substantial fraction of AD subjects are classified as MCI (71 cases), and CN/MCI are frequently misclassified as one another (103 CN predicted as MCI and 113 MCI predicted as CN). This suggests that the depth-space representation captures a gradual transition pattern, but overlaps remain between adjacent stages.

2.7 Method 2: Inductive conformal classification in depth space

Goal. While Method 1 outputs a single diagnostic label, Method 2 aims at providing *uncertainty-aware* predictions. We adopt *inductive conformal classification* in the depth space, which returns a *prediction set* (e.g., {MCI, AD}) guaranteed to contain the true label with a user-chosen probability, under the standard assumption of exchangeability. In particular, for a target miscoverage level $\alpha \in (0, 1)$, the method produces sets $\Gamma_\alpha(x)$ such that the marginal coverage satisfies

$$\mathbb{P}(Y \in \Gamma_\alpha(X)) \geq 1 - \alpha.$$

Train / calibration / test protocol. The dataset is split (stratified by diagnosis) into three disjoint subsets:

- **Training set** ($\approx 60\%$): used to estimate class-related quantities, i.e. the class-wise depth functions;
- **Calibration set** ($\approx 20\%$): used to calibrate conformal p -values via nonconformity scores;
- **Test set** ($\approx 20\%$): used only for final evaluation.

Stratification ensures that CN/MCI/AD proportions are preserved within each split. Importantly, to avoid data leakage, depths for calibration and test subjects are computed using the *training* set as reference, not the full dataset.

Nonconformity scores in depth space. Let $D_y(x)$ denote the depth of subject x with respect to class $y \in \{\text{CN}, \text{MCI}, \text{AD}\}$ (estimated from the training sample of class y). We define the nonconformity score as

$$A(x, y) = -D_y(x). \quad (7)$$

This choice encodes a natural notion of conformity: subjects that are *central* with respect to class y have large depth $D_y(x)$ and therefore small $A(x, y)$ (high conformity), while outlying subjects have small depth and large $A(x, y)$ (low conformity).

Conformal p -values and prediction sets. For each candidate class y , we restrict the calibration set to points whose true label is y and compute their nonconformity scores, obtaining an empirical reference distribution $\{A_i : i \in \text{Cal}(y)\}$. For a test subject x , the conformal p -value for class y is computed as

$$p_y(x) = \frac{\#\{i \in \text{Cal}(y) : A_i \geq A(x, y)\} + 1}{|\text{Cal}(y)| + 1}. \quad (8)$$

Given a miscoverage level α , the conformal prediction set is then

$$\Gamma_\alpha(x) = \{y \in \{\text{CN}, \text{MCI}, \text{AD}\} : p_y(x) > \alpha\}. \quad (9)$$

Smaller α yields larger, more conservative sets (higher coverage), while larger α yields smaller, more decisive sets.

Evaluation metrics. On the held-out test set, we report:

- **Empirical coverage:** the fraction of test subjects whose true label is contained in $\Gamma_\alpha(x)$;
- **Average set size:** the mean cardinality $\mathbb{E}[|\Gamma_\alpha(X)|]$;
- **Set size distribution:** counts of prediction sets of size 0, 1, 2, 3;
- **(Optional) Singleton analysis:** when $|\Gamma_\alpha(x)| = 1$, we treat the output as a point prediction and report singleton rate and accuracy on those cases.

Results ($\alpha = 0.1$). At miscoverage level $\alpha = 0.1$ (target coverage 90%), the conformal classifier achieved:

- **Empirical coverage:** 0.927,
- **Average prediction set size:** 2.69,
- **Singleton rate ($|\Gamma_\alpha(x)| = 1$):** 4.4%.

The distribution of prediction set sizes on the test set was:

$$|\Gamma_\alpha(x)| = 0 : 1 \quad |\Gamma_\alpha(x)| = 1 : 6 \quad |\Gamma_\alpha(x)| = 2 : 28 \quad |\Gamma_\alpha(x)| = 3 : 102.$$

Interpretation. The observed coverage (92.7%) is slightly above the nominal target (90%), indicating that the conformal procedure behaves as expected in terms of marginal validity. However, the predictions are generally not very informative: the average set size is close to the maximum possible value (3), and the vast majority of test subjects receive sets of size 2 or 3. In particular, most subjects are assigned the full set $\{\text{CN}, \text{MCI}, \text{AD}\}$, and only a small fraction (4.4%) receives a singleton label.

This behavior suggests substantial overlap between the three diagnostic groups in the current depth space representation. Under the chosen nonconformity score $A(x, y) = -D_y(x)$, the method often cannot confidently exclude alternative diagnoses, especially for subjects lying in ambiguous regions of the depth space (notably around the MCI group). As a consequence, the conformal predictor remains conservative in order to maintain the desired coverage.

Overall assessment. Inductive conformal classification successfully provides uncertainty-aware predictions with valid coverage guarantees. Nonetheless, under the current depth representation and score definition, its practical decisiveness is limited, as reflected by the high average set size and the very low singleton rate.

2.8 Method 3: k -Nearest Neighbors classification in depth space (DD- k NN)

Overview. In this method, subjects are classified in the 3-dimensional depth space using the k -nearest neighbors (k NN) algorithm. Each subject is represented by the depth triplet

$$z_i = (D_{\text{CN}}(\tilde{x}_i), D_{\text{MCI}}(\tilde{x}_i), D_{\text{AD}}(\tilde{x}_i)) \in \mathbb{R}^3,$$

and class membership is inferred by comparing each subject to nearby training subjects in this depth-based representation. Since k NN is a fully nonparametric method, it does not fit an explicit model; rather, it relies on local neighborhoods in the feature space.

Classification rule in depth space. Given a subject with depth coordinates $z \in \mathbb{R}^3$, we compute its distance to all training subjects in depth space (Euclidean distance in practice). We then identify the k closest points and assign the label by majority vote among their diagnoses. Using an odd value of k helps reduce ties, while larger values of k typically produce smoother (less variable) decision boundaries.

Model selection via stratified 5-fold cross-validation. Performance of k NN depends strongly on the choice of k . Therefore, we tune k using stratified 5-fold cross-validation, preserving CN/MCI/AD proportions in each fold. For each candidate value,

$$k \in \{1, 3, 5, \dots, 355\},$$

we train the classifier on 4 folds and evaluate it on the held-out fold, repeating the procedure across all folds. For each k , we compute the mean cross-validated accuracy and its standard deviation. The optimal value k^* is selected as the one maximizing the mean cross-validated accuracy. A plot of mean accuracy versus k (with ± 1 SD band) is used to assess sensitivity to the neighborhood size and identify possible stable ranges of good-performing k values. Figure ?? reports the mean stratified 5-fold CV accuracy as a function of the number of neighbors k , together with a ± 1 SD band. The selected value $k^* = 33$ is highlighted by the vertical dashed line.

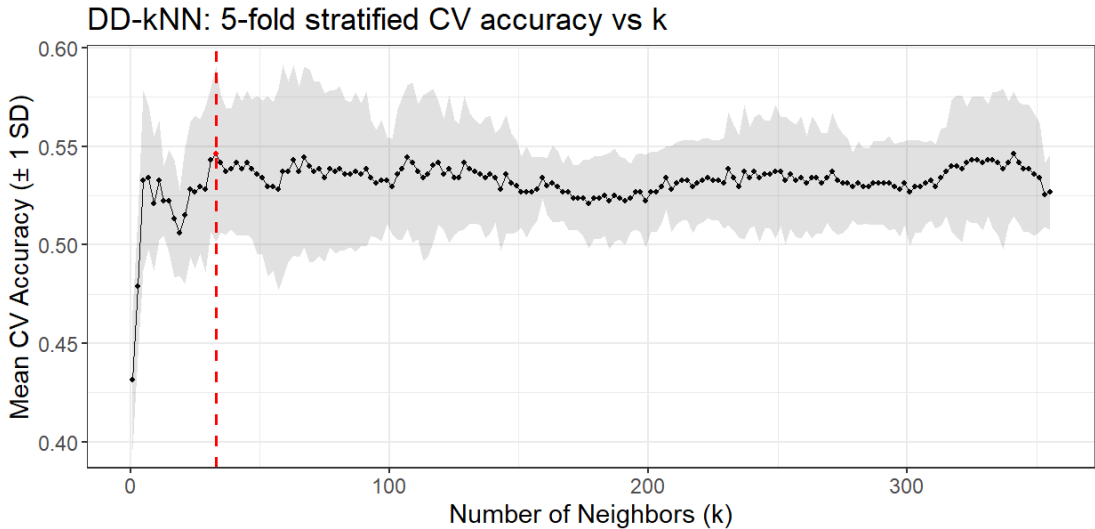


Figure 3: DD- k NN: mean stratified 5-fold cross-validation accuracy as a function of k . The shaded band shows ± 1 standard deviation across folds. The vertical dashed line indicates the selected value $k^* = 33$.

Performance reporting. After selecting k^* , we summarize performance by reporting the cross-validated mean accuracy (and its standard deviation across folds). In addition, a cross-validated confusion matrix can be obtained by pooling predictions from the held-out folds, providing class-wise insight into the most frequent confusions (e.g. CN vs MCI, MCI vs AD).

Results. The value maximizing mean cross-validated accuracy is:

$$k^* = 33.$$

At this choice, the classifier achieves:

- **Mean accuracy (5-fold CV):** 0.5461,
- **Standard deviation across folds:** 0.0444.

Interpretation. DD- k NN leverages the idea that subjects that are close in depth space (i.e., exhibiting similar relative centrality with respect to CN, MCI, and AD) should share the same diagnosis. Because decisions are based on local neighborhoods, the method can capture complex and non-linear class boundaries without imposing a parametric form. The obtained accuracy is comparable to Method 1, suggesting that, under the current depth representation, the main source of difficulty remains the overlap between diagnostic groups in the depth space rather than the specific choice of classifier.

2.9 Method 4: Support Vector Machine (SVM) in depth space

Overview. In this method, classification is performed directly in the 3-dimensional depth space $z_i = (D_{\text{CN}}, D_{\text{MCI}}, D_{\text{AD}})$ using a Support Vector Machine (SVM) with a radial basis function (RBF) kernel. Differently from neighborhood-based methods (e.g. k NN) or depth-based separation rules (e.g. DD- α), the SVM learns a *global* non-linear decision boundary that maximizes the separation margin between classes in the depth space.

In our implementation we use an RBF kernel with fixed hyperparameters:

$$C = 1, \quad \gamma = 0.5.$$

Classification principle. The RBF SVM implicitly maps the 3D depth space into a higher-dimensional feature space and fits separating surfaces that:

- maximize the margin between classes,
- allow non-linear decision boundaries,
- penalize misclassifications through the cost parameter C .

Since the depth space is already low-dimensional and geometrically interpretable, the SVM operates on a compact representation of class centrality rather than on the original high-dimensional atrophy-rate space.

Performance evaluation. Performance is assessed using stratified 5-fold cross-validation: class proportions are preserved in each fold, the model is trained on 4 folds and evaluated on the remaining fold. We report mean accuracy and its standard deviation across folds. Class-wise behavior is summarized by a pooled cross-validated confusion matrix.

Results. Using the RBF SVM with $C = 1$ and $\gamma = 0.5$, stratified 5-fold cross-validation yields:

$$\text{Mean accuracy} = 0.537, \quad \text{SD across folds} = 0.035.$$

The pooled cross-validated confusion matrix is reported in Table 4.

Table 4: Confusion matrix for RBF SVM in depth space (5-fold cross-validation). Rows indicate predicted labels and columns indicate true labels.

Predicted \ True	AD	CN	MCI
AD	7	1	5
CN	2	173	107
MCI	95	101	181

Interpretation. Overall accuracy is moderate ($\approx 53.7\%$), comparable to the other point-prediction methods in depth space. The confusion matrix highlights a consistent difficulty in separating the intermediate MCI group from both extremes. In particular, many AD subjects are predicted as MCI (95 cases), and CN/MCI confusions are frequent (101 CN predicted as MCI and 107 MCI predicted as CN). CN is the most reliably identified class (173 correct classifications), while AD appears to overlap substantially with MCI in the current projection-depth representation. The relatively small standard deviation across folds (≈ 0.035) suggests stable performance across splits.

Overall assessment. The RBF SVM provides a flexible non-linear classifier in depth space. However, under the chosen hyperparameters, its predictive performance remains in the mid-50% range, suggesting that the main limitation is not the classification rule itself but rather the degree of overlap among diagnostic groups (especially involving MCI) in the projection-depth representation.

2.10 Motivation: beyond location-based separation

Following the implementation of the previous classification models in the depth space, the observed predictive performances suggested that the separation among the three diagnostic groups (CN, MCI, AD) cannot be fully explained by differences in *central location*. In particular, despite employing flexible and nonparametric classifiers (DD- α , conformal prediction sets, k NN, and RBF SVM), classification accuracy remained moderate and the three groups exhibited substantial overlap in the depth representation.

A closer inspection of the depth space and of selected regional features indicates that the groups largely coincide in terms of location, while differing more clearly in their *dispersion* and structural variability. Cognitively normal (CN) subjects tend to cluster around a relatively compact configuration, whereas MCI and AD subjects display progressively larger variability and more pronounced deviations from this baseline structure (Figure 4).

These observations motivate a methodological shift. Rather than pursuing further refinements of location-based classifiers, we focus on explicitly quantifying departures from a normative reference distribution. The CN group is therefore adopted as a baseline population, and the subsequent analyses are designed to measure, for each individual, the magnitude and the structural nature of the deviation from this reference.

Under this perspective, disease progression is interpreted not primarily as a uniform shift in central tendency, but as an increasing dispersion and distortion relative to the CN pattern. The following methods are thus constructed to capture baseline departures, with the objective of more effectively characterizing the structural alterations associated with cognitive impairment.

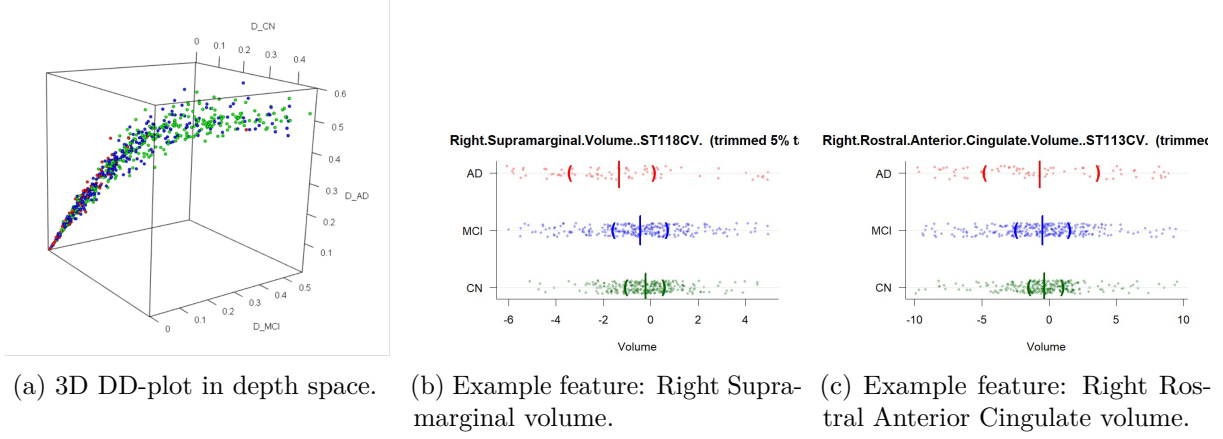


Figure 4: Exploratory diagnostics suggesting that group differences are driven more by dispersion/variability than by a pure location shift. Panel (a) shows the overlap of CN/MCI/AD in depth space, while panels (b)–(c) illustrate two representative regional features where CN appears more compact and MCI/AD display increased spread.

2.11 Dispersion-Based Indices Relative to the CN Baseline

Let X_{ij} denote the value of feature j for subject i . Let m_j and s_j denote respectively the median and MAD of feature j computed within the CN group. We define the robust standardized deviations as

$$Z_{ij} = \frac{X_{ij} - m_j}{s_j}.$$

All dispersion indices introduced below are functions of the deviation vector $Z_i = (Z_{i1}, \dots, Z_{ip})$.

1. Mean Absolute Deviation (L1 Index)

$$T_i^{(1)} = \frac{1}{p} \sum_{j=1}^p |Z_{ij}|.$$

This index measures the average magnitude of deviation from the CN baseline across regions. To stabilize scale and reduce skewness, a logarithmic transformation was applied:

$$\tilde{T}_i^{(1)} = \log(T_i^{(1)}).$$

Cross-validated accuracy: 0.557.

2. Variance of Absolute Deviations

$$T_i^{(2)} = \text{Var}(|Z_{ij}|).$$

This quantity captures the heterogeneity of regional deviations, distinguishing between diffuse and regionally concentrated abnormalities. The log-transformed version was used:

$$\tilde{T}_i^{(2)} = \log(T_i^{(2)}).$$

Cross-validated accuracy: 0.539.

3. Skewness of Absolute Deviations

$$T_i^{(3)} = \text{Skewness}(|Z_{ij}|).$$

This index measures asymmetry in the distribution of deviations across regions and aims to detect the presence of extreme localized alterations. **Cross-validated accuracy:** 0.432.

4. Correlation-Pattern Deviation (Quadratic Form)

Let Σ_{CN} denote the shrinkage covariance matrix of the CN deviation vectors. We define the quadratic deviation score as

$$T_i^{(4)} = Z_i^\top \Sigma_{CN}^{-1} Z_i.$$

This measure incorporates the multivariate correlation structure of the CN baseline and evaluates how strongly the full deviation vector departs from the normative anatomical configuration.

Cross-validated accuracy: 0.600.

2.12 Method 5: Envelope-Based Ordinal Deviation Classifier

In addition to the previously described models, we implemented a dispersion-based classifier built on robust nested diagnostic envelopes derived from the cognitively normal (CN) group. For each brain region, a robust location and scale were estimated using the CN median and median absolute deviation (MAD). Two nested envelopes were then constructed: a tighter CN band and a wider MCI band. For each subject, every regional measurement was classified as (i) within the CN envelope, (ii) within the MCI ring but outside the CN band, or (iii) outside both envelopes. These ordinal deviations were converted into weighted penalties and averaged across regions to produce a single global dispersion score (Term 1). The penalty weights were tuned within each cross-validation fold to maximize between-class separation, and classification thresholds were optimized on the training data. This approach focuses explicitly on dispersion rather than location, quantifying how broadly and severely a subject deviates from normative CN structure. Because it relies on robust statistics (median and MAD), it is resistant to outliers and skewed regional distributions. The method is computationally simple, transparent, and produces a single scalar summary that captures global structural disorganization, which is particularly advantageous in high-dimensional neuroimaging settings where dispersion patterns may carry stronger discriminative information than mean shifts. Total CV accuracy: 0.5560.

Interpretation

Among the considered indices, the correlation-aware quadratic deviation $T^{(4)}$ achieved the highest predictive performance (approximately 0.60). This suggests that departures from the multivariate structural organization of the CN group carry more discriminative information than purely marginal dispersion summaries. However, several structural limitations emerge. The dispersion indices, particularly those based on marginal magnitudes, are strongly correlated and therefore encode largely overlapping information. Furthermore, even the best-performing index yields only moderate classification accuracy. These findings indicate that while baseline-relative dispersion contains the principal discriminative signal in the data, the information available through this representation is intrinsically limited. Most indices effectively reduce the high-dimensional structure to a largely one-dimensional notion of overall deviation from normality. Once this global deviation component is captured, additional transformations contribute only marginal improvements. Consequently, the structural MRI features, when summarized through baseline-relative dispersion measures, appear to encode a restricted amount of separable information across CN, MCI, and AD. This limitation provides a coherent explanation for the moderate ceiling observed across both depth-based and dispersion-based classifiers.

2.13 Stratification by Genotype

To assess whether genetic stratification improves discriminative performance, we repeated the dispersion-based classification analyses after stratifying subjects by genotype. In this setting, models were trained and evaluated within genotype-defined subgroups using the same cross-validation protocol and dispersion indices described above.

Contrary to expectations, genotype stratification did not lead to improved predictive performance. On the contrary, average cross-validated accuracies were systematically lower than those obtained in the pooled analysis. The resulting mean accuracies and standard deviations across folds are reported below:

Table 5: Cross-validated classification accuracy after stratification by genotype.

Dispersion index	Mean accuracy	SD accuracy
$\log(T^{(1)})$ (L1 index)	0.554	0.0566
$T^{(4)}$ (Quadratic form)	0.518	0.0621
$\log(T^{(3)})$ (Skewness of $ Z_{ij} $)	0.517	0.0557
$T^{(3)}$ (Skewness of $ Z_{ij} $)	0.433	0.0662

Interpretation. Genotype stratification leads to a clear deterioration in classification performance across all considered dispersion indices. This degradation is plausibly attributable to the effective reduction in sample size within each genotype subgroup, which increases estimation variance and weakens the stability of both the baseline (CN) reference distribution and the learned classification thresholds.

More substantively, these results suggest that, within the present feature representation, genotype does not induce sufficiently distinct dispersion patterns to compensate for the loss of statistical power incurred by stratification. In other words, genetic heterogeneity does not appear to correspond to markedly different structural deviation profiles in the projection-depth or dispersion spaces considered here. Consequently, incorporating genotype via stratification may be counterproductive for classification in this setting, and pooled analyses yield more reliable and stable performance.

3 Classification based on Functional Depths

3.1 Methodological Framework

Along with atrophy rates, a functional approach was adopted to classify patients. We focused on the Modified Band Depth (MBD), which is particularly suitable for functional datasets like ADNI. This non-parametric measure allows us to quantify the centrality of a subject’s trajectory relative to a group distribution, effectively overcoming the issue of a sparse number of visits—a scenario where traditional smoothing methods, such as splines, often fail to produce stable estimates.

3.1.1 Functional Representation and Age Grid Construction

To compare subject trajectories, regional brain volumes were modeled as functions of age. Since visits occur at irregular intervals, we defined a common age grid of length $L = 50$, spanning the entire age range of the cohort. Individual trajectories were linearly interpolated onto these 50 points, a mathematical prerequisite for the Modified Band Depth (MBD) that ensures all curves are evaluated at the same temporal instances for a synchronized comparison of centrality. Following this interpolation, for each subject i , the MBD was computed for each region j . The final functional depth was then derived as the multivariate average across all regions, defined by the formula $D_k(X_i) = \frac{1}{p} \sum_{j=1}^p \text{MBD}(X_{ij} | G_k)$, where G_k represents the reference set for class k .

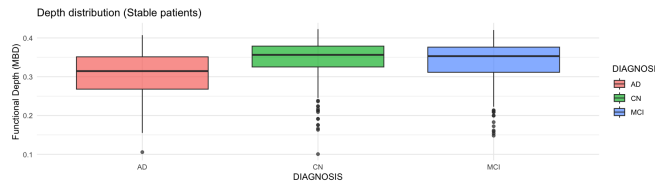
3.1.2 Study Design

Subjects whose diagnosis remained constant (“stable”) were used for model development via 5-fold stratified cross-validation. Subjects who changed diagnosis (“converters”) were reserved for independent external testing (while training on all stable patients), with the target label defined as the final diagnosis recorded at their last visit.

3.1.3 Statistical Inference on Functional Depths

Before classification, we performed Permutational Independence Tests on the depth distributions to validate the discriminative power of the MBD.

- CN vs AD: Highly significant ($p < 0.001$), confirming a clear separation between healthy and pathological trajectories.
- CN vs MCI: Not significant ($p = 0.122$), highlighting a substantial overlap between healthy and prodromal stages when considering volumetric centrality alone.
- MCI vs AD: Highly significant ($p < 0.001$), indicating a detectable shift during the transition to clinical dementia.



3.2 Method 1: Max-Depth Functional Classifier

The maximum depth rule assigns each subject to the class $k \in \{CN, MCI, AD\}$ in which their trajectory appears most central (highest depth value):

$$\hat{y}_i = \arg \max_{k \in \mathcal{G}} \{D_{CN}(X_i), D_{MCI}(X_i), D_{AD}(X_i)\} \quad (10)$$

3.2.1 Three-Class Analysis: 117 Volumes vs. 18 Selected Volumes

We evaluated the impact of feature selection by comparing the full set of 117 regions against a subset of 18 volumes (9 regions, known to be relevant for Alzheimer’s disease in literature: Hippocampus, Amygdala, Entorhinal, Parahippocampal, Fusiform, Lateral Ventricle, Posterior Cingulate, Precuneus, and Middle Temporal).

Performance (Full 117 Volumes):

- Accuracy (Stable CV): 0.449
- Accuracy (Converters): 0.419

Performance (Selected 18 Volumes):

- Accuracy (Stable CV): 0.472 (Balanced: 0.532)
- Accuracy (Converters): 0.362

Interpretation: The substantial drop in accuracy on the converter set suggests that focusing on a limited subset of regions, though biologically relevant, is insufficient.

3.2.2 Binary Analysis: CN vs. AD

Given the permutational results, we restricted the task to the clinical extremes to assess the maximum discriminative potential of functional depth.

Performance:

- Accuracy (Stable CV): 0.711 (Balanced: 0.741)
- Accuracy (Converters, training on all stable): 0.477

Interpretation: The binary model effectively identifies the stable clinical poles, correctly identifying 196 CN and 84 AD subjects.

3.3 Method 2: Logistic Regression with Functional Covariates

The results of the permutation tests revealed that the volumetric centrality of MCI subjects is not significantly different from that of the CN group ($p > 0.05$). This suggests that MCI represents a broad transitional spectrum whose trajectories overlap with both healthy and pathological patterns, blurring the statistical boundaries between classes.

Note on Model Selection: Preliminary experiments using multinomial logistic regression to classify CN, MCI, and AD simultaneously yielded poor accuracy, confirming the difficulty of isolating the MCI group through purely functional metrics. Furthermore, we explored the inclusion of additional anatomical and clinical covariates—specifically **HeadSize** (total supratentorial volume) and the binary clinical variable **DXDEP** (depression)—but found no significant improvement in model performance or discriminant capacity. Consequently, to isolate a clearer diagnostic signal, we adopted a binary logistic regression framework.

This allowed us to evaluate two strategies: one targeting the broad distinction between AD and the rest of the cohort and one focused on the direct contrast between CN and AD. In both specifications, classification is initially performed using a standard threshold of 0.5.

3.3.1 One-vs-Rest Classification: AD vs. non-AD

In this setting, CN and MCI subjects were merged into a single "non-AD" category. This model tests whether the functional and genetic features can discriminate clinical dementia from the rest of the population.

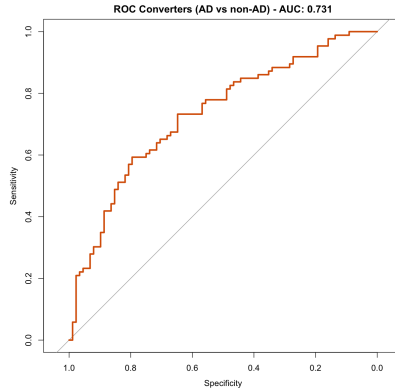
Table 6: Logistic Regression Coefficients (trained on all stable patients): AD vs. non-AD (CN + MCI)

Predictor	Estimate (β)	Odds Ratio (OR)	P-value
Intercept	-3.958	0.019	0.0054
functional_depth	-10.004	< 0.001	$< 2 \cdot 10^{-7}$
age	0.061	1.063	0.0002
GENOTYPE 4/4	2.322	10.196	$< 1 \cdot 10^{-9}$

Model Summary and Performance:

- Accuracy (Stable CV): 0.8476 (Balanced: 0.5690)
- Accuracy (Converters): 0.592 (Standard threshold 0.50)
- AUC (Converters): 0.731
- Youden Index Optimization: Using an optimal threshold of 0.195, the accuracy on Converters significantly improved to 0.695.

Interpretation: Even with the inclusion of the transitional MCI group, `functional_depth` remains a highly significant protective factor. The genetic risk associated with the 4/4 genotype is substantial, increasing the odds of an AD diagnosis by approximately 10 times compared to the reference group. The solid AUC of 0.731 on the converter test set confirms the model's ability to assign higher risk scores to patients progressing toward dementia.



3.3.2 Pairwise Classification: CN vs. AD

To isolate the strongest biological contrast, we restricted the analysis to the clinical extremes, excluding MCI subjects. The probability of Alzheimer's Disease is modeled as:

$$\text{IP(AD)} = \text{logit}^{-1}(\beta_0 + \beta_1 \text{functional_depth} + \beta_2 \text{age} + \beta_3 \text{GENOTYPE}) \quad (11)$$

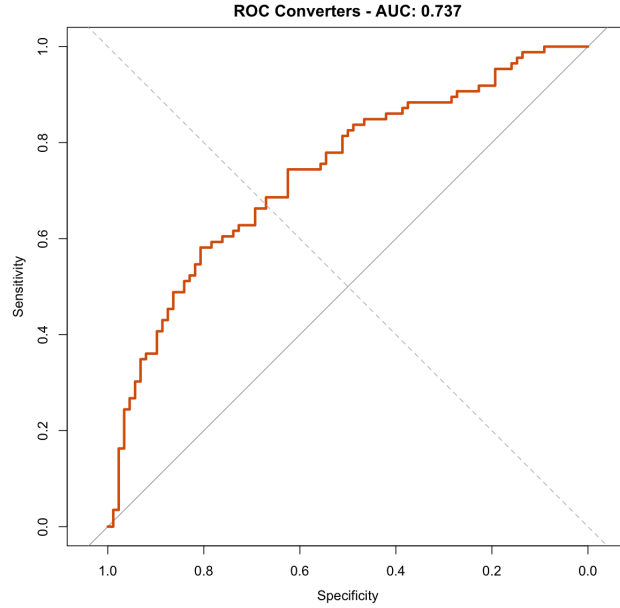
Table 7: Logistic Regression Coefficients (trained on all stable patients): Pairwise (CN vs. AD)

Predictor	Estimate (β)	Odds Ratio (OR)	P-value
Intercept	-3.750	0.023	0.0252
functional_depth	-13.905	< 0.001	$< 3 \cdot 10^{-8}$
age	0.083	1.087	$< 7 \cdot 10^{-5}$
GENOTYPE 4/4	3.592	36.306	$< 3 \cdot 10^{-11}$

Model Summary and Performance:

- Accuracy (Stable CV): 0.804 (Balanced: 0.704)
- Accuracy (Converters): 0.667 (Standard threshold 0.50)
- AUC (Converters): 0.737
- Youden Index Optimization: Shifting the threshold to 0.315 improved the balanced accuracy on the test set to 0.672.

Interpretation: By removing the noise of the transitional MCI stage, the predictors' effects become even more pronounced. The model confirms that **functional_depth** is a protective factor ($OR < 0.001$): maintaining volumetric centrality relative to healthy trajectories drastically reduces disease odds. The risk associated with age increases by 8.7% per year, while the 4/4 genotype emerges as the dominant risk factor, with 36 times higher odds of AD compared to the 3/3 reference group.



Conclusion: The high AUC values (> 0.73) confirm the robust discriminant capacity of the functional logistic regression across different cohort specifications.

4 Functional Envelope-Based Classifier

As a final alternative approach, we implemented a **Functional Envelope-Based Classifier** grounded on pointwise deviations from a normative healthy reference. This method shifts the focus from global typicality to the quantification of regional abnormalities relative to the Cognitively Normal (CN) subjects.

4.1 Envelope Construction and Reference Bands

For each of the $p = 117$ brain regions, CN trajectories (utilizing up to the first four visits per patient) were interpolated onto the common age grid to establish a normative baseline. Using the median trajectory and the Median Absolute Deviation (MAD), we constructed two nested reference envelopes:

- Inner CN Band: Defined as $\text{Median} \pm a \cdot \text{MAD}$, representing the range of typical physiological variation.
- Outer Deviation Band: Defined as $\text{Median} \pm b \cdot \text{MAD}$ ($b > a$), capturing moderate to significant departures from the normative trend.

4.2 Ordinal Penalty Scoring and Optimization

The classifier evaluates a patient’s functional trajectory (baseline-normalized percent change) by comparing it pointwise to these CN envelopes across all regions and time points. To reflect the severity of neurodegeneration, each observation is assigned an ordinal penalty P :

$$P = \begin{cases} 0 & \text{if inside the Inner Band,} \\ w_1 & \text{if between the Inner and Outer bands,} \\ w_2 & \text{if outside the Outer Band,} \end{cases} \quad (12)$$

where the condition $w_2 > w_1$ enforces a logical ordering of severity. The final patient score is the average penalty across all valid region–time points.

Within each cross-validation fold, the weights (w_1, w_2) were optimized to maximize the separation between diagnostic groups, and two thresholds were selected to map the resulting one-dimensional score into the three classes (CN, MCI, AD).

Performance:

- **Accuracy (5-fold CV):** 0.523

Interpretation: This approach successfully translates complex functional abnormalities into an interpretable severity score. The achieved accuracy is competitive with depth-based methods, demonstrating that explicit modeling of departures from a normative envelope captures a significant portion of the discriminative signal in the longitudinal data.

5 Conclusions

This research developed a non-parametric framework to study Alzheimer’s Disease (AD) using longitudinal MRI data. By comparing different methods, we found two main approaches that provide the most useful results for understanding how the disease progresses.

5.1 Final Classification Models

1. Atrophy-Based: Quadratic Deviation Index ($T^{(4)}$)

This model reached the highest accuracy (60%) because it does not look at brain regions in isolation. Instead of just checking if one area is shrinking, it looks at the brain as a whole. It proves that Alzheimer’s is better identified by the breakdown of the teamwork between regions rather than just looking at the volume loss of a single part.

2. Functional-Based: Depth-Logistic Regression (AD vs. non-AD)

This model is very effective (AUC 0.731) because it simplifies years of complex brain changes into a single, easy-to-read score. The results match what doctors see in real life: age and the APOE gene increase the risk, while a high functional depth acts as a shield or a sign of brain health.

5.2 Critical Observations and possible further developments

While the binary model (AD vs. non-AD) yields high accuracy, this result stems from a simplification that bypasses the clinical complexity of the MCI stage. Identifying this intermediate state is crucial for early diagnosis, yet it remains hindered by the "MCI Grey Zone": a mismatch where subjects exhibit accelerated atrophy despite maintaining brain volumes within normal ranges. This overlap creates a statistical ceiling for MRI-only volumetric models. Our results suggest that morphometry alone may be insufficient for absolute diagnostic certainty, highlighting the need for further developments:

- Neural Networks: to capture non-linear structural patterns.
- Expert Validation: involving neurologists to refine "Grey Zone" labels.
- Data enlargement: utilizing larger cohorts and "converters" to model disease dynamics.

The key to resolving the ambiguity of the prodromal phase likely resides in multimodal integration, combining MRI functional depth with CSF or PET biomarkers to move beyond the limitations of purely structural analysis.

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